

Supplemental Material - Proofs for the YFB Model

OpenAI Codex

2026-05-08

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1 Purpose

This document collects several proofs and proof sketches that are useful for explaining the YFB formulation of the SSBMF model.

The focus is on the most relevant structural claims:

1. **train/test consistency of YFB**
2. **decoupling of $q(L_k)$ from survival**
3. **scalar EBNM form of the YFB $q(\tilde{\beta}_k)$ update**
4. **dual-source form of the YFB $q(F_k)$ update**
5. **scale comparison between the original LB supervision path and the YFB supervision path**
6. **sign/orientation invariance**
7. **what can and cannot currently be proved about convergence of the CAVI algorithm**

The last section is intentionally careful. Some convergence statements are well-supported; others would require assumptions or algorithmic cleanup that are not yet fully encoded in the current implementation.

2 Model Setup

The shared genomics model is

$$Y_{ij} \mid L, F, \tau \sim N \left(\sum_{k=1}^K l_{ik} f_{jk}, \tau_j^{-1} \right).$$

Equivalently,

$$Y \approx LF^\top.$$

Under the YFB formulation, the Cox linear predictor is

$$\eta_i = \sum_{k=1}^K (z_F)_{ik} \tilde{\beta}_k, \quad Z_F = YF, \quad (z_F)_{ik} = \sum_{j=1}^p y_{ij} f_{jk}.$$

So the survival model becomes

$$h(t_i \mid y_i, F, \tilde{\beta}) = h_0(t_i) \exp(\eta_i).$$

The mean-field variational family is

$$q(L, F, \tilde{\beta}, \tau) = q(L) q(F) q(\tilde{\beta}) q(\tau).$$

The ELBO is

$$\begin{aligned} \mathcal{L} = & E_q[\log p(Y \mid L, F, \tau)] + E_q[\log p(t, \delta \mid Y, F, \tilde{\beta})] \\ & - \text{KL}\{q(L) \parallel p(L)\} - \text{KL}\{q(F) \parallel p(F)\} \\ & - \text{KL}\{q(\tilde{\beta}) \parallel p(\tilde{\beta})\} - \text{KL}\{q(\tau) \parallel p(\tau)\}. \end{aligned}$$

Proposition 1 (Train/Test Consistency of YFB). *Under the YFB formulation, the prognostic score used at training time and the score used at test time have the same functional form:*

$$\eta_{\text{train}} = Y_{\text{train}} F \tilde{\beta}, \quad \eta_{\text{test}} = Y_{\text{test}} F \tilde{\beta}.$$

By contrast, under LB the training and test predictors are not literally the same object.

Proof. By definition, the YFB Cox predictor is

$$\eta_i = (y_i^\top F) \tilde{\beta}.$$

If the training data matrix is Y_{train} , then stacking all training samples gives

$$\eta_{\text{train}} = Y_{\text{train}} F \tilde{\beta}.$$

For a new test sample y_i^{test} , the model prescribes the same predictor:

$$\eta_i^{\text{test}} = ((y_i^{\text{test}})^\top F) \tilde{\beta}.$$

Stacking test samples gives

$$\eta_{\text{test}} = Y_{\text{test}} F \tilde{\beta}.$$

The formula is identical except for replacing the training matrix with the test matrix. Therefore the predictor form is the same at train and test time. $\square$

Commentary. This is one of the cleanest conceptual motivations for YFB.

In LB, training uses latent L , but prediction on new data cannot use the same latent quantity directly. In YFB, both training and testing use the same observable projection mechanism.

Proposition 2 ($q(L_k)$ Decouples From Survival in YFB). *Under the YFB predictor*

$$\eta = YF\tilde{\beta},$$

the survival term of the ELBO is constant with respect to $q(L_k)$. Therefore the variational update for $q(L_k)$ depends only on the genomics likelihood and the prior on L_k .

Proof. The YFB survival term is

$$E_q[\log p(t, \delta \mid Y, F, \tilde{\beta})].$$

Its arguments are the observed matrix Y , the factor matrix F , and the survival coefficients $\tilde{\beta}$. There is no dependence on L .

Therefore, for any factor k ,

$$\frac{\partial}{\partial q(L_k)} E_q[\log p(t, \delta \mid Y, F, \tilde{\beta})] = 0.$$

So the $q(L_k)$ -dependent part of the ELBO reduces to

$$E_q[\log p(Y \mid L, F, \tau)] - \text{KL}\{q(L_k) \parallel p(L_k)\} + \text{const.}$$

Hence the optimal update for $q(L_k)$ is obtained from the genomics term alone. $\square$

Commentary. This proposition explains why the YFB L update is so much simpler than the LB L update.

In LB, L_k was a supervised object. In YFB, L_k becomes a reconstruction-only object.

Proposition 3 (Scalar EBNM Form of the YFB $q(\tilde{\beta}_k)$ Update). *Under the second-order Cox Taylor surrogate, the variational update for $q(\tilde{\beta}_k)$ reduces to a one-dimensional normal-means problem with*

$$A_{\tilde{\beta},k} = \alpha \sum_i w_i (z_F)_{ik}^2$$

and

$$B_{\tilde{\beta},k} = \alpha \sum_i w_i z_i^{-k} (z_F)_{ik}.$$

Proof. Start from the quadratic Cox surrogate

$$\log p(t, \delta \mid \eta) \approx -\frac{1}{2} \sum_i w_i (z_i - \eta_i)^2 + \text{const.}$$

Now isolate factor k :

$$\eta_i = (z_F)_{ik} \tilde{\beta}_k + \eta_i^{-k}.$$

Define the partial working response

$$z_i^{-k} = z_i - \eta_i^{-k}.$$

Then

$$z_i - \eta_i = z_i^{-k} - (z_F)_{ik} \tilde{\beta}_k.$$

Substitute into the surrogate:

$$-\frac{1}{2} \sum_i w_i \left(z_i^{-k} - (z_F)_{ik} \tilde{\beta}_k \right)^2.$$

Expand the square:

$$\left(z_i^{-k} - (z_F)_{ik} \tilde{\beta}_k \right)^2 = (z_i^{-k})^2 - 2z_i^{-k} (z_F)_{ik} \tilde{\beta}_k + (z_F)_{ik}^2 \tilde{\beta}_k^2.$$

So the $\tilde{\beta}_k$ -dependent part becomes

$$-\frac{1}{2} \sum_i w_i (z_F)_{ik}^2 \tilde{\beta}_k^2 + \sum_i w_i z_i^{-k} (z_F)_{ik} \tilde{\beta}_k + \text{const.}$$

Introducing the mixing weight α , this becomes

$$-\frac{1}{2} A_{\tilde{\beta},k} \tilde{\beta}_k^2 + B_{\tilde{\beta},k} \tilde{\beta}_k + \text{const.},$$

with

$$A_{\tilde{\beta},k} = \alpha \sum_i w_i (z_F)_{ik}^2, \quad B_{\tilde{\beta},k} = \alpha \sum_i w_i z_i^{-k} (z_F)_{ik}.$$

This is exactly the quadratic form underlying a scalar normal-means likelihood with pseudo-observation

$$x_{\tilde{\beta},k} = \frac{B_{\tilde{\beta},k}}{A_{\tilde{\beta},k}}$$

and pseudo-noise

$$s_{\tilde{\beta},k} = \frac{1}{\sqrt{A_{\tilde{\beta},k}}}.$$

Therefore the update is an EBNM problem. □

Commentary. This is one of the most attractive properties of YFB. The survival update becomes regression on an observed factor score, instead of regression on a latent uncertain quantity.

Proposition 4 (Dual-Source Form of the YFB $q(F_k)$ Update). *In the mathematical YFB model, the update for $q(F_k)$ receives one contribution from reconstruction and one contribution from survival. The combined quadratic coefficients are*

$$A_{F,k}(j) = (1 - \alpha) \tau_j \sum_i E[l_{ik}^2] + \alpha E[\tilde{\beta}_k^2] \sum_i w_i y_{ij}^2$$

and

$$B_{F,k}(j) = (1 - \alpha) \tau_j \sum_i r_{k,ij} E[l_{ik}] + \alpha E[\tilde{\beta}_k] \sum_i w_i z_i^{-k} y_{ij}.$$

Proof. **Genomics contribution.**

The factor- k reconstruction term is

$$-\frac{1}{2} \sum_{i,j} \tau_j E[(r_{k,ij} - l_{ik} f_{jk})^2].$$

Expand the square:

$$r_{k,ij}^2 - 2r_{k,ij}l_{ik}f_{jk} + l_{ik}^2f_{jk}^2.$$

Take expectations under mean-field:

$$E[l_{ik}f_{jk}] = E[l_{ik}]E[f_{jk}], \quad E[l_{ik}^2f_{jk}^2] = E[l_{ik}^2]E[f_{jk}^2].$$

Collecting the terms involving a fixed f_{jk} gives

$$-\frac{1}{2} \left(\tau_j E[f_{jk}^2] \sum_i E[l_{ik}^2] - 2\tau_j E[f_{jk}] \sum_i r_{k,ij} E[l_{ik}] \right) + \text{const.}$$

So

$$A_{F,\text{gen},k}(j) = \tau_j \sum_i E[l_{ik}^2], \quad B_{F,\text{gen},k}(j) = \tau_j \sum_i r_{k,ij} E[l_{ik}].$$

Survival contribution.

Now isolate the dependence of the YFB predictor on f_{jk} . Since

$$(z_F)_{ik} = \sum_j y_{ij} f_{jk},$$

the predictor contains the term

$$y_{ij} f_{jk} \tilde{\beta}_k.$$

So, holding all other terms fixed,

$$\eta_i = y_{ij} f_{jk} \tilde{\beta}_k + \eta_i^{-j,k}.$$

Using the quadratic Cox surrogate,

$$-\frac{1}{2} \sum_i w_i (z_i^{-k} - y_{ij} f_{jk} \tilde{\beta}_k)^2.$$

Expanding the square gives

$$-\frac{1}{2} \sum_i w_i \left[(z_i^{-k})^2 - 2z_i^{-k} y_{ij} f_{jk} \tilde{\beta}_k + y_{ij}^2 f_{jk}^2 \tilde{\beta}_k^2 \right].$$

The terms depending on f_{jk} are

$$-\frac{1}{2} \left(E[f_{jk}^2] E[\tilde{\beta}_k^2] \sum_i w_i y_{ij}^2 - 2E[f_{jk}] E[\tilde{\beta}_k] \sum_i w_i z_i^{-k} y_{ij} \right) + \text{const.}$$

Therefore

$$A_{F,\text{surv},k}(j) = E[\tilde{\beta}_k^2] \sum_i w_i y_{ij}^2, \quad B_{F,\text{surv},k}(j) = E[\tilde{\beta}_k] \sum_i w_i z_i^{-k} y_{ij}.$$

Combined update.

Combine the two sources with mixing weight α :

$$A_{F,k}(j) = (1 - \alpha) A_{F,\text{gen},k}(j) + \alpha A_{F,\text{surv},k}(j),$$

$$B_{F,k}(j) = (1 - \alpha)B_{F,\text{gen},k}(j) + \alpha B_{F,\text{surv},k}(j).$$

Thus the update is again a normal-means / EBNM problem with

$$x_{F,jk} = \frac{B_{F,k}(j)}{A_{F,k}(j)}, \quad s_{F,jk} = \frac{1}{\sqrt{A_{F,k}(j)}}.$$

□

Commentary. This is arguably the main theoretical proposition of YFB.

It formalizes the claim that YFB moves supervision from L to F .

Proposition 5 (Scale Comparison Between LB and YFB). *Assume there exist positive constants $c_\tau, C_\tau, c_f, C_f, c_l, C_l, c_w, C_w, c_y, C_y, c_\beta, C_\beta$, independent of n and p , such that for the factor k under consideration:*

$$\begin{aligned} 0 < c_\tau \leq \tau_j \leq C_\tau, \quad 0 < c_f \leq E[f_{jk}^2] \leq C_f, \\ 0 < c_l \leq E[l_{ik}^2] \leq C_l, \quad 0 < c_w \leq w_i \leq C_w, \\ 0 < c_y \leq y_{ij}^2 \leq C_y, \quad 0 < c_\beta \leq E[\tilde{\beta}_k^2], E[\beta_k^2] \leq C_\beta \end{aligned}$$

for all relevant i, j . Then:

1. the LB ratio $A_{L,\text{surv},i}/A_{L,\text{gen}}$ is $O(p^{-1})$ uniformly in i ;
2. the intended YFB ratio $A_{F,\text{surv},k}(j)/A_{F,\text{gen},k}(j)$ is $O(1)$ uniformly in j .

Proof. **LB.**

In LB, the L -update precision has

$$A_{L,\text{gen}} = \sum_{j=1}^p \tau_j E[f_{jk}^2].$$

Using the lower and upper bounds,

$$\sum_{j=1}^p c_\tau c_f \leq A_{L,\text{gen}} \leq \sum_{j=1}^p C_\tau C_f,$$

so

$$p c_\tau c_f \leq A_{L,\text{gen}} \leq p C_\tau C_f.$$

Hence $A_{L,\text{gen}} = \Theta(p)$.

The survival precision term is

$$A_{L,\text{surv},i} = w_i E[\beta_k^2].$$

Again by the bounds,

$$c_w c_\beta \leq A_{L,\text{surv},i} \leq C_w C_\beta$$

uniformly in i . Hence $A_{L,\text{surv},i} = \Theta(1)$.

Therefore

$$\frac{c_w c_\beta}{p C_\tau C_f} \leq \frac{A_{L,\text{surv},i}}{A_{L,\text{gen}}} \leq \frac{C_w C_\beta}{p c_\tau c_f},$$

uniformly in i . This proves

$$\frac{A_{L,\text{surv},i}}{A_{L,\text{gen}}} = \Theta(p^{-1}),$$

hence in particular it is $O(p^{-1})$.

YFB.

In the intended YFB F update, the genomics precision is

$$A_{F,\text{gen},k}(j) = \tau_j \sum_{i=1}^n E[l_{ik}^2].$$

Using the bounds,

$$\tau_j n c_l \leq A_{F,\text{gen},k}(j) \leq \tau_j n C_l,$$

and since $c_\tau \leq \tau_j \leq C_\tau$,

$$n c_\tau c_l \leq A_{F,\text{gen},k}(j) \leq n C_\tau C_l.$$

Hence $A_{F,\text{gen},k}(j) = \Theta(n)$ uniformly in j .

The survival precision is

$$A_{F,\text{surv},k}(j) = E[\tilde{\beta}_k^2] \sum_{i=1}^n w_i y_{ij}^2.$$

Using the bounds,

$$E[\tilde{\beta}_k^2] n c_w c_y \leq A_{F,\text{surv},k}(j) \leq E[\tilde{\beta}_k^2] n C_w C_y,$$

and since $c_\beta \leq E[\tilde{\beta}_k^2] \leq C_\beta$,

$$n c_\beta c_w c_y \leq A_{F,\text{surv},k}(j) \leq n C_\beta C_w C_y.$$

Hence $A_{F,\text{surv},k}(j) = \Theta(n)$ uniformly in j .

Therefore

$$\frac{c_\beta c_w c_y}{C_\tau C_l} \leq \frac{A_{F,\text{surv},k}(j)}{A_{F,\text{gen},k}(j)} \leq \frac{C_\beta C_w C_y}{c_\tau c_l},$$

uniformly in j . Thus

$$\frac{A_{F,\text{surv},k}(j)}{A_{F,\text{gen},k}(j)} = \Theta(1),$$

hence in particular it is $O(1)$. □

Commentary. This is not a full asymptotic theorem, but it is the right structural scaling argument.

It explains why YFB was attractive in the first place: it relocates supervision onto a quantity where the survival and genomics terms operate on comparable sample-level scale.

Proposition 6 (Sign/Orientation Invariance). *For either LB or YFB, a coupled sign flip of a factor score and its corresponding survival coefficient leaves the linear predictor unchanged. Therefore the sign of the prognostic axis is not uniquely identified by the factorization itself.*

Proof. **LB.**

The predictor is

$$\eta = L\beta.$$

If factor k and coefficient β_k are both sign-flipped:

$$l_{\cdot k} \rightarrow -l_{\cdot k}, \quad \beta_k \rightarrow -\beta_k,$$

then

$$(-l_{\cdot k})(-\beta_k) = l_{\cdot k}\beta_k.$$

So the predictor is unchanged.

YFB.

The predictor is

$$\eta = Z_F \tilde{\beta}, \quad Z_F = YF.$$

If $f_{\cdot k} \rightarrow -f_{\cdot k}$, then $Z_{F, \cdot k} \rightarrow -Z_{F, \cdot k}$.

If we also flip $\tilde{\beta}_k \rightarrow -\tilde{\beta}_k$, then

$$(-Z_{F, \cdot k})(-\tilde{\beta}_k) = Z_{F, \cdot k}\tilde{\beta}_k.$$

So the predictor is again unchanged. □

Commentary. This proposition is why the post-fit sign/orientation issue is not purely a bug. It is a real identifiability feature of the model family.

Proposition 7 (Surrogate-Level Coordinate-Ascent Monotonicity). *Fix one outer-iteration Taylor expansion of the Cox likelihood. Suppose:*

1. *the surrogate objective is used as the optimization target for that outer iteration,*
2. *each block update solves its EBNM subproblem exactly,*
3. *all other blocks are held fixed during that coordinate update.*

Then each block update is non-decreasing in the surrogate objective.

Proof. Let the fixed surrogate objective at a given outer iteration be denoted by

$$\mathcal{S}(q_L, q_F, q_{\tilde{\beta}}, q_{\tau}).$$

Write the four variational blocks abstractly as

$$\mathbf{q} = (q_1, q_2, q_3, q_4),$$

where $q_1 = q_L$, $q_2 = q_F$, $q_3 = q_{\tilde{\beta}}$, and $q_4 = q_{\tau}$. Let $\mathbf{q}^{(m,r)}$ denote the state after outer iteration m and after the first r block updates of the inner loop, with $r \in \{0, 1, 2, 3, 4\}$. Thus $\mathbf{q}^{(m,0)}$ is the state before the inner loop and $\mathbf{q}^{(m,4)}$ is the state after all four blocks have been updated.

Fix a block index $b \in \{1, 2, 3, 4\}$. By assumption, the b -th block update is the exact solution of

$$q_b^{(m,b)} \in \arg \max_{q_b} \mathcal{S}(q_1^{(m,b-1)}, \dots, q_{b-1}^{(m,b-1)}, q_b, q_{b+1}^{(m,b-1)}, \dots, q_4^{(m,b-1)}).$$

Therefore, evaluating the surrogate objective at the maximizing block value cannot produce a smaller value than evaluating it at the previous block value $q_b^{(m,b-1)}$. Hence

$$\mathcal{S}(\mathbf{q}^{(m,b)}) \geq \mathcal{S}(\mathbf{q}^{(m,b-1)}).$$

Since this argument holds for every block b , each block update is non-decreasing in the fixed surrogate objective. $\square$

Corollary 1 (Convergence of Surrogate Objective Values). *Assume, in addition to the hypotheses of Proposition 7, that:*

1. *the fixed surrogate objective is bounded above,*
2. *every block update is well-defined,*
3. *the sequence of iterates remains in a region where the block maximizers exist.*

Then the sequence of surrogate-objective values generated by the inner coordinate-ascent loop is monotone non-decreasing and convergent.

Proof. Let

$$s_r = \mathcal{S}(\mathbf{q}^{(m,r)}), \quad r = 0, 1, 2, 3, 4, \dots$$

denote the surrogate-objective values generated by the inner coordinate-ascent loop for the fixed outer-iteration surrogate. By Proposition 7,

$$s_{r+1} \geq s_r \quad \text{for all } r.$$

Thus $\{s_r\}_{r \geq 0}$ is monotone non-decreasing. By assumption, the surrogate objective is bounded above on the region visited by the iterates, so there exists $M < \infty$ such that

$$s_r \leq M \quad \text{for all } r.$$

Therefore $\{s_r\}$ is a bounded monotone sequence in $\mathbb{R}$. By the monotone convergence theorem for real sequences, $\{s_r\}$ converges. $\square$

Commentary. This corollary concerns convergence of objective values for a *fixed* Taylor surrogate, not yet the full outer algorithm.

Proposition 8 (Limits of a Full Global Convergence Claim). *A full global convergence theorem for the current practical YFB fitting pipeline does not follow directly from Proposition 7 or Corollary 1.*

Proof. The current practical algorithm differs from textbook block coordinate ascent on a single fixed objective in several respects.

First, the Cox quadratic surrogate is recomputed at each outer iteration, so the objective optimized by the inner loop changes with the current predictor.

Second, the post-fit sign correction is applied after the main iterative fitting loop rather than encoded as an explicit block of the optimization problem.

Third, the implemented YFB update for F uses the stabilized choice $\alpha_F = 0$ rather than the fully coupled theoretical dual-source update.

Because of these features, a theorem establishing global convergence of the complete practical pipeline would require a more detailed treatment of the outer-majorization step, the interaction between refreshed surrogates and inner solves, and the role of the orientation rule. Those ingredients are not supplied by the present propositions alone. $\square$

Commentary. The appropriate formal claim at the current stage is surrogate-level coordinate-ascent monotonicity, not global optimality or a guaranteed iteration bound for the full practical algorithm.

3 Final Summary

The most useful proofs for YFB right now are:

1. YFB uses the same predictor form at train and test time.
2. $q(L_k)$ decouples completely from survival.
3. $q(\tilde{\beta}_k)$ is a scalar EBNM update on projection scores.
4. $q(F_k)$ is theoretically dual-source.
5. YFB fixes the structural scale mismatch that hurt the original LB supervision path.
6. Sign ambiguity is a genuine identifiability issue, not just a coding mistake.
7. The implemented CAVI algorithm supports a meaningful surrogate-level coordinate-ascent convergence argument, but a stronger full global theorem would require additional formal cleanup.

That set of results gives a solid theoretical backbone for discussing the YFB model rigorously without overstating what has or has not yet been proved.